PATTERN EXPLORERS

A Class VI Maths Activity Workbook

Inspired by ‘Patterns in Mathematics’ — Chapter 1

Name: _____

Date started: _____

Date finished: _____

Welcome, Pattern Explorer! Numbers and shapes love to repeat, grow, and surprise us. In this workbook you'll hunt for number patterns, build shapes out of dots, meet a famous number sequence found in flowers and shells, and even invent a pattern of your own. Grab a pencil, some colours if you have them, and let's get exploring!

1. Number Detective — crack the missing-number codes
2. Shape Train — continue the shape pattern
3. Triangular Number Towers — build with dots
4. Square Number Gardens — colour the grids
5. The Growing Rabbit Sequence (Virahanka/Fibonacci numbers)
6. Polygon Scavenger Hunt
7. Pattern Word Problems
8. Design Your Own Pattern — free creativity page
Answer Key

Activity 1
Number Detective

Each row follows a secret rule. Find the rule and fill in the missing numbers.

A	2	5	8	11	___	17

Rule: add ____

B	1	2	4	8	___	32

Rule: ____

C	50	45	40	___	30	25

Rule: subtract ____

D	1	4	9	16	___	36

Rule: these are _____ numbers

E	1	1	2	3	5	___	13

Rule: add the _____ two numbers

Bonus: Pick your favourite row above and write the next THREE numbers in the pattern on the back of this page.

Activity 2
Shape Train

Circle, square, circle, square... patterns aren't just for numbers! Draw the next 3 shapes in each train.

● ■ ● ■ ● ■ → draw what comes next: ____ ____ ____

▲ ▲ ● ▲ ▲ ● → draw what comes next: ____ ____ ____

★ ● ● ★ ● ● → draw what comes next: ____ ____ ____

Now invent your own shape train using at least 2 different shapes, and ask a family member to guess what comes next!

Activity 3

Triangular Number Towers

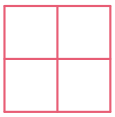
Triangular numbers are made by stacking rows of dots: 1, then 1+2, then 1+2+3, and so on. Draw the dots for the towers below, then fill in the total.

Tower	Rows of dots	Total dots (triangular number)
T1		
T2		
T3		
T4		

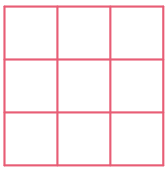
Do you see a pattern in how much each tower grows compared to the one before it? Write it here:

Activity 4
Square Number Gardens

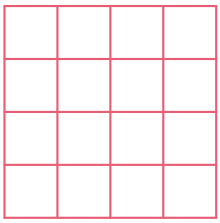
A square number is made by arranging dots (or flowers!) into a perfect square grid. Colour in the grids to match each square number, then write the multiplication.



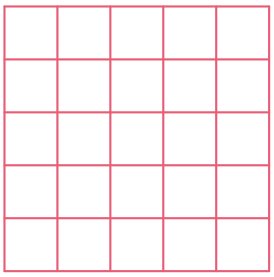
2 × 2 = _____



3 × 3 = _____



4 × 4 = _____



5 × 5 = _____

Activity 5

The Growing Rabbit Sequence

This famous pattern was described by the Indian mathematician Virahanka long before it was named 'Fibonacci' in Europe. Each number is the sum of the two numbers before it. You can find it in sunflower seed spirals, pinecones, and snail shells!

1	**1**	**2**	**3**					

Fill in the 5 missing boxes above.

Now go on a hunt around your home or garden: can you spot a spiral pattern in a flower, a shell, a pinecone, or even a fern leaf? Draw or describe what you found:

Activity 6
Polygon Scavenger Hunt

A regular polygon has all sides and all angles equal. Look around your house for objects shaped like the polygons below, and write down what you find.

Shape	Name	Number of sides	Object I found
Triangle	Triangle	3	
Square	Square	4	
Pentagon	Pentagon	5	
Hexagon	Hexagon	6	
Octagon	Octagon	8	

Activity 7
Pattern Word Problems

Use what you've learned to solve these.

1. Riya is stacking cups in a triangle. Row 1 has 1 cup, row 2 has 2 cups, row 3 has 3 cups, and so on. How many cups does she need for a tower with 6 rows?

2. A garden path is laid with tiles in this pattern: red, red, blue, red, red, blue, ... What colour will the 20th tile be?

3. A plant doubles its number of leaves every week. It starts with 3 leaves. How many leaves will it have after 4 weeks?

4. Look at the numbers 1, 4, 9, 16, 25. What is the NEXT square number, and what is 10 × 10 equal to?

Activity 8
Design Your Own Pattern

Use the space below to invent a brand-new number pattern OR shape pattern. Write the rule, then challenge someone at home to figure it out!

My pattern's rule:

Answer Key (for grown-ups)

Activity 1 — Number Detective

• A: 14 (add 3)
• B: 16 (double each time)
• C: 35 (subtract 5)
• D: 25 (square numbers: 1,4,9,16,25,36)
• E: 8 (Fibonacci-style: add previous two)

Activity 3 — Triangular Numbers

• T1 = 1, T2 = 3, T3 = 6, T4 = 10
• Each tower adds one more row than the last (the growth itself increases by 1 each time).

Activity 4 — Square Numbers

• 2×2=4, 3×3=9, 4×4=16, 5×5=25

Activity 5 — Growing Rabbit Sequence

• 1, 1, 2, 3, 5, 8, 13, 21, 34 (each number = sum of previous two)

Activity 6 — Polygon sides

• Triangle 3, Square 4, Pentagon 5, Hexagon 6, Octagon 8

Activity 7 — Word Problems

• 1) 1+2+3+4+5+6 = 21 cups
• 2) Pattern repeats every 3 tiles (R,R,B); 20 ÷ 3 = 6 remainder 2, so the 20th tile matches the 2nd in the pattern → Red
• 3) 3 → 6 → 12 → 24 → 48 leaves after 4 weeks
• 4) Next square number is 36; 10×10 = 100

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